Christian Scherer

Full-Stack Systems Architect | Cloud Infrastructure, Zero Trust Networking, Automation Design, Systems Integration & Software Development
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PROFILE

Full-stack and AI systems engineer building software, automation, and AI tooling that runs enterprises. Over eight years at Burns & McDonnell modernizing Microsoft 365 and ServiceNow at enterprise scale, now bridging IBM i platforms to Azure at Murphy-Hoffman Company. n8n and LLM pipelines, RouteLLM routing, OpenCode/DeepSeek agentic development, multi-agent systems, local LLMs, and self-hosted speech (Whisper STT, Kokoro/Piper/eSpeak TTS) — all deployed governance-first under Zero Trust, with disposable-container verification gates, a container control plane, disaster-recovery rebuild tooling, self-healing automation (Kuma/ntfy alerts, watchdogs, nightly appdata backups), and E2E-encrypted remote access and an encrypted AI operations console with all-session cost analytics. Strong across Python, TypeScript, .NET, React, FastAPI, Docker, PostgreSQL, and CI/CD. Adds a read-only Sentinel observer, local FTS5 recall, a four-lab model council, and a supervised remote-desktop gateway.

SKILLS

Full-Stack Development — Advanced: Python, TypeScript, React, Next.js, .NET, C#, FastAPI, SQL, PostgreSQL, Vite, Vercel
AI / LLM Engineering — Advanced: LLMs, RAG / retrieval, Prompt engineering, Multi-agent systems, RouteLLM, OpenAI-compatible API, Ollama, OpenCode, DeepSeek, Agentic development, Whisper STT, Text-to-Speech, Speech-to-Text, Kokoro, Piper, eSpeak, episodic memory, Multi-model council, Sentinel observability, FTS5 recall
Automation Orchestration — Expert: n8n, NocoDB, JobSpy, PowerShell, ServiceNow, Workflow Automation, pytest, Browserless, PyJWT, Socket proxy, Cron scheduling, Health watchdogs
Cloud & Infrastructure — Advanced: Azure, Docker, Linux, Cloudflare Zero Trust, Terraform, CI/CD, GitHub Actions, Neon Postgres, Render, Vercel, Kasm, Docker SDK, Cloudflare Access, Disaster Recovery, Git submodules, Uptime Kuma, ntfy, Healthchecks, Self-healing automation, Cron/systemd, SQLite backup, OpenCode API, Encrypted chat, Cost analytics, Kuma monitoring, Remote-desktop gateway
Enterprise Platforms — Expert: Microsoft 365, Exchange Online, Entra ID, Intune, Active Directory, SCCM
Legacy & Integration — Advanced: IBM i, RPG/CL, System i, REST APIs, OAuth, Azure tokens

EXPERIENCE

Programmer/Analyst — Murphy-Hoffman Company (MHC Kenworth) (2025-10–Present)
Developed and enhanced business applications on the IBM i platform using RPG and CL programming; integrated modern technologies like Azure, Python, and .NET with legacy systems to boost efficiency; collaborated with internal stakeholders to design and implement solutions that streamline operations.
Staff Software Developer — Burns & McDonnell (2023-03–2025-09)
Designed, coded, and tested software for business challenges; led architecture and modernization incl. QTS modernization for Azure hosting; built secure REST API integrations (OAuth, Azure tokens); automated IT processes (licensing, storage archiving, data normalization) saving hundreds of admin hours annually; mentored developers via code reviews.
Staff IT System Specialist — Burns & McDonnell (2023-01–2023-03)
Directed ServiceNow architectural changes: new catalog modules and automation streamlining request handling for thousands of users; integrated workflows with Azure-hosted services incl. ESRI backup automation; documented technical decisions with governance compliance.
Assistant IT System Specialist — Burns & McDonnell (2022-07–2023-01)
Developed and implemented new application modules and service catalog workflows within the Now platform, enhancing system efficiency; analyzed business requirements and documented technical solutions to improve system functionality; troubleshot technical issues throughout the Software Development Life Cycle, ensuring smooth operation.

Assistant IT Applications Specialist — Burns & McDonnell (2021-08–2022-07)
Troubleshot critical issues and assisted with problem resolution for customers; developed and maintained knowledge base articles to enhance user support; researched, verified, and documented product defects to improve overall product quality.
End User Support Technician — Burns & McDonnell (2017-06–2021-08)
Deployed computer software systems to engineers locally and in the field; assisted engineers with software and hardware issues occurring with their computer systems.
Laptop Repair Technician — Heartland Services Inc (2016-08–2017-06)
Diagnosed and repaired computers and tablets for Panasonic, including hardware and software fixes; conducted board and LCD replacements and other hardware fixes to ensure optimal device performance; collaborated with team members to streamline repair processes and improve efficiency in diagnosing issues.

PROJECTS

OP-E Console — Encrypted AI Operations Console
End-to-end encrypted AI operations console: OpenCode/DeepSeek sessions, resume, screenshot attachments, all-session cost/token analytics, live server health, isolated workspace, Kuma monitoring/systemd restart, and verified backups, behind Cloudflare Access. Includes the Mind observer feed and episodic memory. Adds a read-only Sentinel observer, local FTS5 recall, a four-lab model council, and a supervised remote-desktop gateway.
Command Deck — Container Control Plane & Tool Console
Config-driven control plane and 43-tool console over 43 containers: live Docker discovery, health probes, host/container stats, guarded actions; FastAPI + Docker SDK behind a scoped socket proxy with Cloudflare Access JWT.
Self-Healing Stack Automation
Cron-driven operations layer: CPU/health watchdogs, Sonarr queue janitor, qBittorrent executable blocking, Kuma-to-ntfy alerts, and nightly SQLite/Postgres appdata backups mirrored across drives.
OP-E Voice — Local Speech Service
Self-hosted STT/TTS for an agent interface: faster-whisper transcription and Kokoro/Piper/eSpeak synthesis with per-emotion prosody. No cloud speech APIs.
Command Center — Disaster-Recovery Rebuild Repo
Sanitized disaster-recovery repo for a multi-service server: services pinned as submodules, host-detecting bootstrap, backup/restore tooling, and rebuild runbooks.
Secure-Term — Encrypted Browser Terminal
E2E-encrypted long-poll browser terminal that keeps terminal content opaque to edge and TLS-inspection proxies: Python stdlib HTTP server, AES-GCM (WebCrypto), paramiko SSH into persistent tmux sessions, constant-time auth, no inbound ports.
Agent-Test — Disposable-Container Verification Pipeline
Self-contained CI harness that runs static analysis, unit/API/UI tests, and hardcoded-value scans in ephemeral containers without touching live services; gates merges on reproducible results.
Job Search Automation Dashboard — https://jobdashboard.christianscherer.dev
Automated job-search pipeline: daily JobSpy scraping, a verified evidence-backed skill inventory, LLM-based fit scoring, tailored resume PDF + cover letter generation, and company research with per-job study guides and interview prep (n8n, NocoDB, FastAPI, Python).
RouteLLM CLI Development Pipeline — https://routellm.abacus.ai
Integrated RouteLLM, an OpenAI-compatible LLM routing gateway, as the single endpoint in the dev pipeline for agentic and automation workflows.
Local LLM / AI Agent Orchestration
Homelab stack orchestrating local LLMs and multi-agent workflows with Ollama; containerized, isolated, accessed via Zero Trust.
IBM i to Azure Integration Bridges
Bridging core IBM i (RPG/CL) business systems to Azure with Python/.NET integration layers, modernizing legacy platforms without rewriting.
Personal Portfolio Platform — https://christianscherer.dev
React/TypeScript portfolio site with an ASP.NET Core API and Postgres backend, including an admin portal for project management.
Track-Folio

Full-stack application-tracking solution web app.

EDUCATION

Southern New Hampshire University — Computer Science (2020–2022)
Coastal Carolina Community College — Computer Programming (2013–2015)
University of North Carolina at Charlotte — Computer Science (2012–2013)